

# MERN STACK

DAY – 17

14 July 2026

## React Router

React Router is a library that enables client-side routing in React applications. It lets users navigate between pages without reloading the entire website.

Example:

- / → Home page
- /about → About page
- /contact → Contact page

### 1. BrowserRouter

BrowserRouter is the top-level component that enables routing using the browser's History API. It wraps your entire application so React Router can manage URLs.

#### Syntax

```
import { BrowserRouter } from "react-router-dom";

function App() {

  return (

    <BrowserRouter>

      {/* All routes go here */}

    </BrowserRouter>

  );

}
```

### 2. Routes

Routes is a container that holds all the Route components. It renders the first route that matches the current URL.

#### Syntax

```
<Routes>

  <Route path="/" element={<Home />} />

</Routes>
```

```
<Route path="/about" element={<About />} />
</Routes>
```

### 3. Route

A Route maps a URL path to a React component.

#### Syntax

```
<Route path="/about" element={<About />} />
```

#### Attributes

| Prop    | Description          |
|---------|----------------------|
| path    | URL path             |
| element | Component to display |

#### Link

Link is a component provided by React Router that is used to create navigation links between different pages in a React application.

It works like an HTML `<a>` (anchor) tag, but it does not reload the entire page. Instead, it updates the URL and allows React Router to display the required component using client-side routing.

#### Syntax:

```
<Link to="/about">About</Link>
```

#### useState

useState is a React Hook that allows functional components to store and manage state (data that can change over time).

Before Hooks, state was mainly used inside class components. With useState, functional components can also create and update state.

#### Import Syntax

To use useState, import it from React:

```
import { useState } from "react";
```

#### Basic Syntax

```
const [state, setState] = useState(initialValue);
```

| Part         | Meaning                           |
|--------------|-----------------------------------|
| state        | Current value stored in state     |
| setState     | Function used to update the state |
| initialValue | Starting value of the state       |

```
import { useState } from "react";

export const Us=()=>{
  const [name,setname]=useState("Aditi")
  const [na,setna]=useState([]);
  const ad=()=>{
    setna([...na,name])
  }
  return(
    <>
    <input type="text" onChange={(e)={setname(e.target.value)}}/>
    <button onClick={ad}>ADD</button>
    {
      na.map((a)=>
        <li>{a}</li>
      )
    }
    </>
  );
}
```

ADITI TANGRI  
ASHMEEN KAUR  
ANJALI KUMARI